

E-HANK: what the welfare numbers are, and what they are not

Working note for Boris and Francesca

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Boris asked whether the CEV numbers in the paper are subject to the well-known bias of first-order welfare evaluation (Kim and Kim, 2003; Benigno and Woodford, 2005). Short answer: the numbers *are* first-order approximations to welfare, so the concern is legitimate. They are not the stochastic case in which first-order evaluation is known to give spurious rankings, but the shock is large and the adoption margin is convex, so the size of the neglected second-order terms is an empirical question, which Section 4 settles with the nonlinear solver: levels are overstated by up to 10–15% at the headline shock, every policy ordering in the paper survives, and one sign claim (the crisis-only greening dividend) must be dropped. This note states precisely what the code computes, why it is the leading-order object, where it may fail, and what the check shows.

1. What `welfare.py` computes

Utilitarian welfare of discount-factor type i along a perfect-foresight transition indexed by shock size ε is

$$W_i(\varepsilon) = \sum_{t \geq 0} \beta_i^t \left[\int u(c_t(x; \varepsilon)) d\Lambda_{i,t}(x; \varepsilon) - v(n_t(\varepsilon)) \right], \quad v(n) = \varphi \frac{n^{1+1/\nu}}{1+1/\nu}, \quad (1)$$

where $x = (d, e, a)$ is the household state, $\Lambda_{i,t}$ the type- i distribution, and $v(n)$ the labour disutility implied by the union's wage-setting condition (uniform hours, so it is common to all households).¹

The code evaluates the transition part of (??) from the *linear* impulse response. `UTIL_i` is a het-output ($u(c)$ aggregated over the distribution); its linear IRF, returned by `solve_impulse_linear`, is by construction the first derivative of aggregate flow utility with respect to the shock,

$$d\text{UTIL}_{i,t} = \int u'(\bar{c}) \partial_\varepsilon c_t d\bar{\Lambda}_i + \int u(\bar{c}) \partial_\varepsilon \Lambda_{i,t}, \quad (2)$$

so that

$$dW_i = \underbrace{\frac{\text{UTIL}_i^{\text{pre}} - \text{UTIL}_i^{\text{base}} - dv^{\text{lvl}}}{1 - \beta_i}}_{\text{standing gap between steady states (exact)}} + \underbrace{\sum_t \beta_i^t (d\text{UTIL}_{i,t} - v'(\bar{n}) dn_t)}_{\text{crisis transition (first order in } \varepsilon)}, \quad (3)$$

and, with log utility, $\chi_i = \exp((1 - \beta_i) dW_i) - 1$, the permanent proportional consumption change that delivers dW_i . The mapping from dW_i to χ_i is exact; dW_i itself is $O(\varepsilon)$ with the $O(\varepsilon^2)$ terms dropped. The standing term (ETS steady state versus baseline steady state) comes from two nonlinear steady-state solves and is exact.

So the honest description of every CEV in the paper is: *the first-order Taylor expansion of welfare in the shock size, around a distorted steady state, along a deterministic transition*. What is dropped is

$$\frac{1}{2} \varepsilon^2 W_i'' = \frac{1}{2} \varepsilon^2 \sum_t \beta_i^t \left[\int (u''(\bar{c})(\partial_\varepsilon c_t)^2 + u'(\bar{c}) \partial_\varepsilon^2 c_t) d\bar{\Lambda}_i + 2 \int u'(\bar{c}) \partial_\varepsilon c_t \partial_\varepsilon \Lambda_{i,t} + \int u(\bar{c}) \partial_\varepsilon^2 \Lambda_{i,t} - v''(dn_t)^2 - v' \partial_\varepsilon^2 n_t \right]. \quad (4)$$

¹With `ghh_prefs=0` preferences are separable; $v(n)$ never enters the household's own problem and is added ex post from the aggregate n path.

2. Why this is nevertheless the leading-order object

The Kim–Kim “spurious welfare reversal” and the Benigno–Woodford LQ apparatus address the *stochastic* problem: expected welfare $\mathbb{E}W(\varepsilon)$ under recurring shocks. There $\mathbb{E}[\varepsilon W'] = 0$ by certainty equivalence, so the welfare effect of a policy is entirely second order, $\frac{1}{2}\sigma^2\mathbb{E}[W'']$, and a first-order solution gets W'' wrong because it omits $\partial_\varepsilon^2 c$ in (??). Rankings based on it can flip. Benigno and Woodford show that when the steady state is distorted, even the LQ objective requires a second-order approximation of the constraints.

Our exercise is different in one respect that matters: it is a single deterministic MIT transition. Then $\varepsilon W'_i \neq 0$ generically—the steady state is not Pareto-efficient (incomplete markets, markups, sticky wages), and the shock is a terms-of-trade income loss with first-order distributional incidence—so the first-order term is the dominant one and (??) captures it. The linear IRF is exactly the first-order perturbation solution (Boppart, Krusell and Mitman, 2018; Auclert, Bardóczy, Rognlie and Straub, 2021), so this is the standard object one obtains from a linearised HANK; it is not the object that Kim–Kim show to be spurious. For $\varepsilon \rightarrow 0$ the levels and rankings in the paper are correct to leading order.

3. Where it can fail: the shock is not small

Two facts in the paper qualify the “leading order” defence.

- The headline price shock doubles the world price ($\varepsilon = 1$). This is not a neighbourhood of the steady state.
- Appendix B documents that the adoption share is strongly convex in the shock: the nonlinear-to-linear ratio of the peak D^G response rises from 1.16 at $\varepsilon = 1/8$ to 2.04 at $\varepsilon = 3/4$, and the nonlinear solver does not converge at $\varepsilon = 1$. Aggregate output is essentially linear (ratio 1.00–1.01), which is why we have described the linear solution as reliable “for aggregate and consumption-based statistics”—a claim about allocations, not about welfare.

Orders of magnitude for the terms in (??). The curvature term $\frac{1}{2}u''(\bar{c})(dc)^2$ with consumption falling about 4–5% per quarter is roughly 2% of the first-order term—benign. The dangerous terms are those involving $\partial_\varepsilon^2 c$ and $\partial_\varepsilon^2 \Lambda$ through the adoption margin (switching costs paid, insulation gained, selection of who switches), which are exactly the ones the linear solution understates by up to a factor two.

The consequence is a distinction between the paper’s claims:

Comparison (price shock, CEV%)	gap	status at first order
transfer -0.82 vs cap -1.15	0.33 pp	probably robust; to be verified
flat transfer -0.32 vs Slutsky -0.82	0.50 pp	probably robust; to be verified
cap sign flip with supply elasticity	~ 0.7 pp	probably robust; to be verified
ex-ante ETS -1.852 vs laissez-faire -1.871	0.02 pp	inside the approximation error
greening dividend by persistence, $+0.01 \rightarrow -0.09$ pp	≤ 0.1 pp	sign not established at first order

The first three rest on gaps that are themselves first order and large relative to any plausible second-order correction. The last two do not: “ex ante barely insures” is safe as a statement of equality, but “the greening dividend turns negative” (Section 6.4) is a sign claim resting on differences of a few basis points and should not be made from a first-order welfare calculation. Either we show it survives the nonlinear check or we restate it as “zero to first order”.

4. The nonlinear check

The sequence-space toolbox solves the nonlinear perfect-foresight transition (`solve_impulse_nonlinear`: Newton on the aggregate unknowns, household block re-solved exactly at every iteration). Along that path `UTIL_i` is the exact aggregate of $u(c)$ over the exact distribution, so $(?)$ is evaluated without approximation. `run_nl_cev.py` does this for laissez-faire, cap, Slutsky transfer, flat transfer and the ex-ante ETS at $\varepsilon \in \{0.25, 0.5, 0.75\}$ (the headline $\varepsilon = 1$ does not converge; the cap does not converge at 0.75).

ε	Scenario	CEV% (linear)	CEV% (nonlinear)	NL/L	impat.	middle	patient
0.25	Laissez-faire	-0.471	-0.463	0.98	0.98	0.98	0.99
0.25	Ex-post cap	-0.288	-0.285	0.99	0.99	0.99	0.99
0.25	Slutsky transfer	-0.205	-0.198	0.97	0.95	0.97	0.99
0.25	Flat transfer	-0.079	-0.076	0.96	1.16	0.94	0.99
0.25	Ex-ante ETS	-0.434	-0.428	0.99	0.99	0.99	0.99
0.5	Laissez-faire	-0.940	-0.900	0.96	0.95	0.96	0.97
0.5	Ex-post cap	-0.575	-0.565	0.98	0.97	0.98	0.99
0.5	Slutsky transfer	-0.409	-0.377	0.92	0.88	0.92	0.96
0.5	Flat transfer	-0.159	-0.139	0.87	1.46	0.83	0.97
0.5	Ex-ante ETS	-0.909	-0.875	0.96	0.97	0.97	0.95
0.75	Laissez-faire	-1.407	-1.301	0.93	0.92	0.93	0.93
0.75	Slutsky transfer	-0.613	-0.524	0.86	0.79	0.85	0.91
0.75	Flat transfer	-0.238	-0.175	0.74	1.89	0.65	0.92
0.75	Ex-ante ETS	-1.382	-1.283	0.93	0.94	0.93	0.90

Four things follow.

1. *Levels.* The first-order CEV overstates the welfare loss, and the gap grows with the shock: 1–2% at $\varepsilon = 0.25$, 4–13% at 0.5, 7–26% at 0.75. The direction is the one the convexity of the adoption margin predicts: the true switching wave is larger (nonlinear/linear peak D^G of 1.3, 1.7, 2.0 at the three sizes) and cushions more. Extrapolating, the paper’s levels at $\varepsilon = 1$ are upper bounds on the losses, by something like 10–15% for laissez-faire and the cap and more for the transfers. They should be reported with that caveat.
2. *Ordering.* Flat > Slutsky > cap > ETS > laissez-faire at every size where all scenarios converge, linear and nonlinear alike. The first-order gaps barely move: Slutsky–cap +0.083 → +0.087 pp at 0.25 and +0.166 → +0.188 at 0.5; flat–Slutsky +0.125 → +0.122 and +0.250 → +0.238. The paper’s welfare rankings stand.
3. *Ex ante vs laissez-faire.* The gap is +0.037 → +0.034 pp at 0.25, +0.031 → +0.025 at 0.5, +0.025 → +0.018 at 0.75: a few basis points, equal to the standing gain of the ETS steady state (+0.043) minus a crisis-time dividend that is small and shrinking in ε . “Ex ante barely insures” is robust. The sign of the crisis-only dividend is not something a first-order calculation can establish, and Section 6.4 should be rephrased accordingly (“zero to first order, and if anything falling with the size and persistence of the shock”).
4. *Distribution.* The one place the linear solution errs in the other direction is the impatient type under the flat transfer: NL/L of 1.16, 1.46, 1.89. The linear solution understates the gain of high-MPC households when the transfer relaxes a binding liquidity constraint (asymmetry of u'). The distributional result of Section 5.5 (flat dominates energy-indexed targeting) is reinforced, not weakened, by the nonlinear check.

On precedents: we do not know of a HANK paper solved with the sequence-space Jacobian that derives a second-order welfare objective. Bhandari, Evans, Golosov and Sargent (2021)

do second-order (small-noise) welfare in HANK, outside SSJ, for optimal Ramsey policy under recurring shocks—the stochastic case. Dávila and Schaab (2025) provide the exact decomposition of heterogeneous-agent welfare changes and would be the natural frame if we go further than the ordering check.

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